

ÖZYEĞİN UNIVERSITY
Graduate School of Science and Engineering

**Design and Experimental Characterization of
a QCL-Based QEPAS Sensor for Trace
Nitrous Oxide (N₂O) Detection**

by

Burak Arayıt

A thesis submitted to
the Graduate School of Science and Engineering
in partial fulfillment of the requirements for the degree of

Master of Science
in
Electrical and Electronics Engineering

Özyeğin University
Month Year

Approval

This thesis titled “**Design and Experimental Characterization of a QCL-Based QEPAS Sensor for Trace Nitrous Oxide (N₂O) Detection**” and submitted by **Burak Arayıt** in partial fulfillment of the requirements for the degree of Master of Science in Electrical and Electronics Engineering has been examined and approved by the committee below.

Prof. Dr. Name Surname (Advisor)

Prof. Dr. Name Surname

Prof. Dr. Name Surname

Date of approval: _____

Declaration

I hereby declare that this thesis is my own work, that it has been produced in accordance with academic rules and ethical conduct, and that it has not been submitted, in whole or in part, for any other degree or qualification. I have fully cited and referenced all material and results that are not original to this work.

Name, Last name: Burak Arayıt

Signature: _____

Abstract

Write a self-contained summary of your thesis here (typically 150–300 words): the problem you address, your approach/method, the key results, and what they mean. The abstract should read on its own, without reference to the rest of the document, and should avoid citations, figures, and undefined abbreviations.

Keywords: quartz-enhanced photoacoustic spectroscopy (QEPAS), nitrous oxide, quantum cascade laser, pulsed QCL, trace gas detection, amplitude modulation, quartz tuning fork

Acknowledgements

I would like to thank my advisor, Prof./Assoc. Prof./Asst. Prof. Advisor Name, for their guidance and support throughout this work. **[TODO: Personalize this section.]**

Contents

Approval	i
Declaration	ii
Abstract	iii
Acknowledgements	iv
1 Introduction	1
1.1 Motivation	1
1.2 Problem Statement and Objectives	1
1.3 Contributions	2
1.4 Thesis Outline	2
2 Background and Related Work	3
2.1 Photoacoustic Spectroscopy	3
2.2 Quartz-Enhanced Photoacoustic Spectroscopy (QEPAS)	3
2.3 Detection Scheme: Amplitude Modulation with a Pulsed QCL	3
2.4 Figures of Merit	4
2.5 Spectroscopy of Nitrous Oxide	4
2.6 Related Work	4
3 Sensor Design and Experimental Setup	5
3.1 System Overview	5
3.2 Pulsed Quantum Cascade Laser	5
3.3 Amplitude-Modulation Detection Scheme	5
3.4 Acoustic Detection Module	6
3.5 Optical Alignment and Power Budget	6
4 Results and Discussion	7
4.1 Tuning-Fork / ADM Characterization	7
4.2 Lock-In Amplifier Characterization	7
4.3 QCL Characterization	7
4.4 Optical Power Budget	7
4.5 Nitrous Oxide Detection	7
4.6 Sensitivity and Stability	8

4.7 Discussion	8
5 Conclusion and Future Work	9
5.1 Conclusion	9
5.2 Future Work	9
References	10
A Supplementary Material	11

List of Figures

3.1	Schematic of the pulsed-QCL QEPAS sensor.	5
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List of Tables

3.1	Main system parameters. [TODO: Replace with your measured values.]	6
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Chapter 1

Introduction

1.1 Motivation

Nitrous oxide (N_2O) is one of the most important long-lived greenhouse gases. Its global-warming potential is roughly 270 times that of CO_2 over a 100-year horizon, and it is currently among the dominant ozone-depleting gases emitted by human activity. **[TODO: Add exact GWP figure and cite (e.g. IPCC AR6) and the dominant-ODS claim (Ravishankara *et al.*, 2009).]** The main anthropogenic sources are agriculture and soil management, combustion, and industrial processes, while N_2O also appears in medical and breath-analysis contexts. Monitoring N_2O at trace (sub-ppm) levels therefore matters for climate science, environmental regulation, and health.

Detecting such low concentrations demands sensors that are sensitive, selective, fast, and ideally compact. Laser absorption spectroscopy meets these needs, and among its variants *quartz-enhanced photoacoustic spectroscopy* (QEPAS) is particularly attractive: it uses a millimetre-scale quartz tuning fork (QTF) as a sharply resonant acoustic transducer instead of a microphone, giving an immunity to environmental acoustic noise, a very small required gas volume, and a wide dynamic range [1], [2]. Combined with a quantum cascade laser (QCL) emitting on a strong N_2O absorption band in the mid-infrared, QEPAS is well suited to compact N_2O sensing.

The broader aim of this line of work is the optical detection of trace volatile organic compounds (VOCs), such as those emitted by plants. N_2O is used here as a well-characterized reference analyte: it has strong, well-documented absorption lines that make it ideal for building and benchmarking the sensor before moving to more demanding targets.

1.2 Problem Statement and Objectives

This thesis concerns the design, construction, and experimental characterization of a QCL-based QEPAS sensor, operated with a *pulsed* QCL in amplitude-modulation mode, and its validation on N_2O . The specific objectives are to:

- assemble and align the optical, acoustic, and electronic subsystems of the QEPAS sensor;

- characterize the key components — the quartz tuning fork / acoustic detection module (ADM), the lock-in detection chain, and the pulsed QCL;
- establish an optical power budget for the beam-delivery path;
- select a suitable N_2O absorption line and demonstrate detection, including a concentration calibration; and
- quantify the sensor's performance: sensitivity (normalized noise-equivalent absorption), minimum detectable concentration, and stability via Allan-deviation analysis.

1.3 Contributions

The main contributions of this thesis are:

- a working pulsed-QCL QEPAS sensor, designed and built as part of this work;
- a thorough experimental characterization of its components and overall detection performance; and
- a validated N_2O measurement that serves as a foundation for future trace-VOC detection.

1.4 Thesis Outline

Chapter 2 reviews photoacoustic spectroscopy, the QEPAS technique, and the spectroscopy of N_2O . Chapter 3 describes the sensor design, the pulsed amplitude-modulation detection scheme, and the experimental setup. Chapter 4 presents the component characterization and N_2O detection results. Chapter 5 concludes and outlines future work toward trace-VOC sensing.

Chapter 2

Background and Related Work

This chapter provides the background needed to follow the rest of the thesis: the principle of photoacoustic spectroscopy, the QEPAS technique and its transducer, the detection scheme used with a pulsed laser, the figures of merit used to compare sensors, and the spectroscopy of N_2O .

2.1 Photoacoustic Spectroscopy

Explain the photoacoustic effect: modulated optical absorption heats the gas periodically, generating a pressure (acoustic) wave that a transducer detects. Note that the signal is proportional to the absorbed optical power, so it scales with laser power and with analyte concentration.

2.2 Quartz-Enhanced Photoacoustic Spectroscopy (QEPAS)

Describe how QEPAS replaces the microphone with a quartz tuning fork acting as a resonant piezoelectric transducer [1], optionally with an acoustic micro-resonator (the ADM). Discuss the high quality factor Q , the narrow resonance at f_0 , and why this gives strong noise immunity and a small sampling volume [2].

2.3 Detection Scheme: Amplitude Modulation with a Pulsed QCL

For a continuous-wave laser, wavelength modulation with second-harmonic (2f) lock-in detection is common. With a *pulsed* QCL, as used in this thesis, the laser intensity is instead modulated at the tuning-fork resonance f_0 (amplitude modulation), and the photoacoustic signal is recovered with a lock-in amplifier at the fundamental (1f). The absorption line is scanned by slowly tuning the laser wavelength via temperature and/or drive current. Explain the trade-offs of pulsed operation (e.g. spectral broadening, duty cycle) relative to CW.

2.4 Figures of Merit

Define the metrics used later: signal-to-noise ratio, the normalized noise-equivalent absorption (NNEA, in $\text{cm}^{-1} \text{ W Hz}^{-1/2}$), minimum detectable concentration, and the use of the Allan deviation to find the optimal integration time.

2.5 Spectroscopy of Nitrous Oxide

Summarize the N_2O absorption bands and justify the chosen line. The strong ν_3 fundamental near $4.5 \mu\text{m}$ (**[TODO: confirm: your QCL center wavelength / target line, e.g. around 2200/cm]**) is a typical choice. Discuss line selection using the HITRAN database [3] and the main spectral interferents (CO_2 , CO , H_2O).

2.6 Related Work

Briefly survey prior QEPAS sensors — especially N_2O and other QCL-based systems — and state how this work relates to and differs from them.

Chapter 3

Sensor Design and Experimental Setup

This chapter describes the QEPAS sensor built in this work: its optical, acoustic, and electronic subsystems, the pulsed amplitude-modulation detection scheme, the alignment procedure, and the optical power budget.

3.1 System Overview

Give the high-level architecture and a block diagram (Figure 3.1): pulsed QCL → beam-delivery optics → ADM (gas cell + QTF) → transimpedance preamplifier → lock-in amplifier → acquisition. Table 3.1 lists the main system parameters.

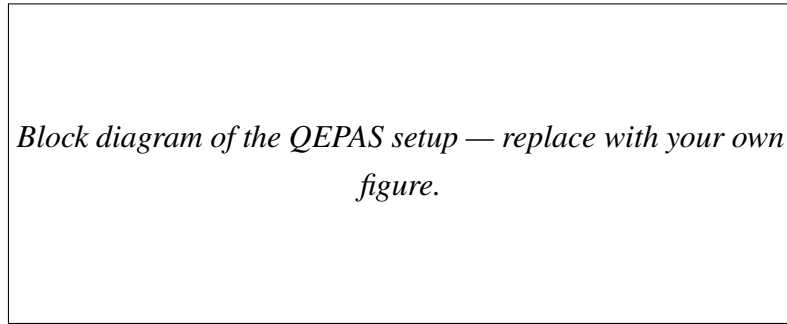


Figure 3.1. Schematic of the pulsed-QCL QEPAS sensor.

3.2 Pulsed Quantum Cascade Laser

Describe the QCL: center wavelength / target N₂O line (**[TODO: fill in]**), pulse parameters (pulse width, repetition rate, duty cycle), the current driver, and temperature (TEC) control. Explain how the wavelength is scanned across the absorption line via temperature/current tuning.

3.3 Amplitude-Modulation Detection Scheme

Explain how the pulse train is modulated at the tuning-fork resonance f_0 and how the photoacoustic signal is recovered with the lock-in amplifier at $1f$. State the modulation frequency, lock-in time constant, and filter settings used.

3.4 Acoustic Detection Module

Describe the ADM: the quartz tuning fork and acoustic micro-resonator geometry, the gas cell, and the gas-handling system (flow, pressure, N₂O source and dilution / calibration mixtures).

3.5 Optical Alignment and Power Budget

Describe the alignment procedure for coupling the QCL beam through the ADM, and present the optical power budget — transmission and losses along the path (Table 3.1 or a dedicated table). This quantifies how much power reaches the QTF, which sets the achievable signal.

Table 3.1. Main system parameters. **[TODO: Replace with your measured values.]**

Parameter	Symbol	Value
QCL center wavelength	λ	4.5 μm (TBD)
Pulse repetition rate	f_{rep}	TBD
QTF resonance frequency	f_0	TBD
QTF quality factor	Q	TBD
Lock-in time constant	τ	TBD

Chapter 4

Results and Discussion

This chapter reports the experimental characterization of the sensor components and the N₂O detection results, followed by a discussion of the achieved performance and its limitations.

4.1 Tuning-Fork / ADM Characterization

Report the measured resonance frequency f_0 and quality factor Q of the QTF (e.g. from a resonance sweep), and any dependence on gas pressure. These set the operating modulation frequency and the integration bandwidth.

4.2 Lock-In Amplifier Characterization

Report the lock-in characterization: response vs. time constant / filter settings, noise floor, and the resulting detection bandwidth.

4.3 QCL Characterization

Report the QCL output power and its wavelength tuning vs. temperature/current, which maps drive settings onto the N₂O absorption line.

4.4 Optical Power Budget

Present the measured transmission and power losses along the beam path and the power delivered to the QTF, comparing with the design budget from Section 3.5.

4.5 Nitrous Oxide Detection

Show the photoacoustic signal as the laser is scanned across the selected N₂O line, and the concentration calibration (signal vs. N₂O concentration), commenting on linearity.

4.6 Sensitivity and Stability

Report the achieved sensitivity — signal-to-noise ratio, normalized noise-equivalent absorption (NNEA), and minimum detectable concentration — and the Allan-deviation analysis identifying the optimal integration time.

4.7 Discussion

Interpret the results, compare with values reported in the literature, and discuss limitations and sources of uncertainty (e.g. pulsed-mode spectral broadening, alignment sensitivity, interferences).

Chapter 5

Conclusion and Future Work

5.1 Conclusion

Summarize what was accomplished: the design and construction of a pulsed-QCL QEPAS sensor, the characterization of its tuning fork / ADM, lock-in chain, and QCL, the optical power budget, and the validated detection of N₂O with its measured sensitivity and stability. Restate the headline performance numbers and why they matter.

5.2 Future Work

Outline the next steps. The primary direction is extending the platform from N₂O to the detection of trace volatile organic compounds (VOCs), such as plant-emitted VOCs, with the goal of a research publication. Discuss the improvements this may require (e.g. line selection for the target VOC, higher sensitivity, interference handling, and calibration).

References

- [1] A. A. Kosterev, Y. A. Bakhirkin, R. F. Curl, and F. K. Tittel, “Quartz-enhanced photoacoustic spectroscopy,” *Optics Letters*, vol. 27, no. 17, pp. 1902–1904, 2002. DOI: [10.1364/OL.27.001902](https://doi.org/10.1364/OL.27.001902)
- [2] P. Patimisco, G. Scamarcio, F. K. Tittel, and V. Spagnolo, “Quartz-enhanced photoacoustic spectroscopy: A review,” *Sensors*, vol. 14, no. 4, pp. 6165–6206, 2014. DOI: [10.3390/s140406165](https://doi.org/10.3390/s140406165)
- [3] I. E. Gordon et al., “The HITRAN2020 molecular spectroscopic database,” *Journal of Quantitative Spectroscopy and Radiative Transfer*, vol. 277, p. 107 949, 2022. DOI: [10.1016/j.jqsrt.2021.107949](https://doi.org/10.1016/j.jqsrt.2021.107949)

Appendix A

Supplementary Material

Use appendices for material that supports the thesis but would interrupt the flow of the main text: long derivations, extra plots, code listings, datasheets, or detailed tables.

Each `\chapter` after `\appendix` is lettered automatically (Appendix A, Appendix B, ...).